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From self-descriptions (SD) to self-recommendations (SR): Evolving Gaia-X for the future European economy

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ABSTRACT

The European decentralized-organized data-infrastructure ecosystem Gaia-X faces the challenge of initiating, maintaining, and intensifying data exchange between data providers and data consumers in all its Gaia-X data exchange domains to contribute significantly to the future viability of the European economy. The overall success of Gaia-X would enhance the innovative power and competitiveness of European companies and reduce their dependence on American and Chinese platform companies. This paper analyzes how the Big Data era affects the design of centralized-organized and decentralized-organized platforms and applies the findings to Gaia-X. It proposes to extend the Self-Descriptions (SD) already firmly embedded in Gaia-X with so-called Self-Recommendations (SR) because, in this way, the decentralized-organized data-infrastructure ecosystem Gaia-X can better express its advantages over centralized-organized platform companies.

1. Introduction

The success of the European decentralized-organized data-infrastructure ecosystem Gaia-X depends on the success of data exchange between Gaia-X providers, which are organized in numerous different Gaia-X domains, and Gaia-X consumers so that innovations emerge and the competitiveness of the organizations organized in Gaia-X is sustainably strengthened. The need for data exchange to take place on the decentralized-organized Gaia-X data infrastructure ecosystem is intensified by the fact that, given the size of the project about the integration and organization of all industrial sectors of the European Union, the future competitiveness of Europe as a business location also depends on a data exchange taking place on Gaia-X. This means there is a risk that the European data exchange will not occur on Gaia-X. In concrete terms, there is a danger that European companies will become irresolvable dependent on American and Chinese centralized platform companies. For this reason, this paper analyzes from a data-theoretic perspective how the characteristics of the Big Data era affect centralized and decentralized-organized platforms. It applies these findings to choosing the most target-oriented design principles of data exchange on Gaia-X. It is recommended that the unilateral use of self-descriptions (SD) on Gaia-X is not sufficient for establishing and maintaining a data exchange, which should, therefore, be supplemented by self-recommendations (SR). This is because Self-Recommendations (SR) extend the power of self-descriptions (SD) in that providers give not only self-declared information about the data but self-declared insights about the results of their data offerings, thereby supporting data consumers in their decision-making in a goal-oriented way. First, we analyze how data properties are fundamentally affected in the context of the Big Data era. It will be demonstrated that the increase in the volume of available data is accompanied by an increase in the self-expressivity of the data. Subsequently, it will be shown that the process of increasing the self-expressivity of data proceeds in combination with the process of datafication, which comprises a prospective and a retractive aspect. It will then be argued how the use of artificial intelligence adds the element of prediction to the process of datafication with its prospective and retractive aspects and what impact these data-evoked and data-driven transformations have on centralized-organized and decentralized-organized platforms. It will emerge that a capacity advantage on the part of centralized-organized platforms is offset by an acceleration advantage on the part of decentralized-organized platforms. Finally, the results of the analysis are applied to Gaia-X by showing that Gaia-X, as a decentralized-organized data-infrastructure ecosystem, can only exploit its acceleration advantage if the concept of self-descriptions (SD), which is already firmly anchored in Gaia-X, is extended to include the novel concept of self-recommendations (SR). Chapter two presents the

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methodology of the paper, followed by a discussion of the Fourth Industrial Revolution in chapter three. Chapter four then explains the characteristics of ecosystems and chapter five provides an insight into the big data era. Chapter six contains the logical conclusion of the extension from self-descriptions to self-commendations, building on the previous chapters. Chapter seven contains the contribution to the literature and the practical implications of this work, and chapter eight concludes.

2. Theory building methodology

There is a great need for new theoretical approaches in information system studies. This is because information systems are constantly changing in different areas of society, which is why it is becoming increasingly essential to describe these phenomena of change theoretically and then to support them scientifically by designing new theoretical foundations in line with the state of the art. (Burton-Jones et al., 2021, p.301) This paper follows the call for the development of novel powerful theories by developing a component of a theory of decentralized ecosystems using the research methodology of the theory-generative practice of analogization as established by Hassan et al., 2019 and Rivard, 2021 for Information System Studies. The research method of theory building based on generative practices works by rearranging components of a theory, such as concepts and assertions, in order to use this reconstruction of the components to define the framework for the construction of a new theory for a specific phenomenon. The research method of generative practices starts one step before the derivation and testing of propositions because it is about the original production of novel ideas and concepts, which can only be subjected to empirical tests in a second step. (Hassan et al., 2019, p.200) It is about the original production of an explanation for the nature of relationships between concepts within a delimited number of assumptions and constraints in relation to a specific phenomenon. (Rivard, 2021, p.317) For this, it is important for researchers to engage deeply with the specific phenomenon in order to arrive at creative insights that enable a completely new way of looking at the specific phenomenon. (Gregory & Henfridsson, 2021, p.1509) One of the most powerful generative practices is analogization, which is a process in which concepts from one specific and often non-adjacent phenomenon area are successfully transferred to another phenomenon area by establishing or initially only assuming that both phenomenon areas behave comparably in this specific respect. (Hassan et al., 2019, p.207) This paper has used the generative practice of analogization by applying common economic concepts of the existence of a well-founded information situation to the successful conditions of data exchange processes in decentralized organized ecosystems. The criterion for measuring the success of theorizing lies in the value of the concepts that emerge from the process of theorizing, which often not only reorganize existing theories but also reconstitute them in a new way. (Hassan et al., 2019, p.212) In addition, different types of theorizing can be distinguished in the context of information systems studies. (Gregor, 2006) This paper develops components of a theory of the theory-type design and action. The theory-type design and action aims at the actual intervention of the developed findings in real contexts in order to validate the accuracy of the predictions and the stability of the components of the theory. This allows further instructions for practical action to be derived because the intervention has already proven itself in practice. (Mueller & Urbach, 2017, p.362) One of the core arguments of this paper is that the introduction of self-recommendations (SR) as an extension of self-descriptions (SD) will have a positive effect on the incentivization of data exchange processes based on a number of theoretical findings so that it is a theory-generative approach in the sense of the theory type design and action, which can also be empirically tested and verified in the form of intervention in existing systems. If this intervention proves to be fruitful, then this would also be proof of the fruitfulness of the concept of self-recommendations (SR) and, thus, the seal of approval for the concept.

3. The fourth industrial revolution

The global economy is undergoing the Fourth Industrial Revolution, driven by the improvement and diffusion of new types of cross-industry key technologies (Mukhopadhyay et al., 2024). It will determine the future success or failure of national economies. As a result of this sustainable global transformation of economic activity and the associated competitive options, visions and roadmaps are currently being developed worldwide on how to create and expand competitive advantages in terms of the creation and use of the key technologies of the Fourth Industrial Revolution in the global competitive environment of nations. (Chou, 2018, p.107) All industrial revolutions share the common characteristic that they are technology-induced and affect the entire economy. The first industrial revolution was triggered in Great Britain with the invention of the steam engine, which greatly simplified the production of goods in factories, resulting in productivity gains. (Li et al., 2017, p.626) In particular, this meant that physically strenuous work in factories could be done by machines. The Second Industrial Revolution was due to the invention of electricity, which allowed for more sophisticated forms of division of labor and further revolutionized the production of goods. The Third Industrial Revolution, in turn, was brought about by the integration of information technologies to control and optimize production processes. (Xu et al., 2018, p. 91) The key technologies of big data, artificial intelligence, and robotics drive the Fourth Industrial Revolution. It is foreseeable that the improvement and large-scale diffusion of these technologies will transform the global economic environment of the 21st century in a profound and lasting way comparable to what has already happened in the context of the three previous industrial revolutions. These key technologies, undergoing continuous improvement, also build on the technological achievements of the Third Industrial Revolution. Moreover, they are interrelated in terms of increasing their performance. (Chen, 2017, p.114) This is because the Internet of Things will generate ever more significant amounts of data, which will perfect artificial intelligence applications, improving the Internet of Things itself in a further iterative step. (Chou, 2018, p.117) This increasing interdependence of key technologies in the context of the Fourth Industrial Revolution will also be particularly present in the smart factories of the future because the numerous components of the value chain will communicate intelligently with each other in the sense of the Internet of Things and thus generate a wealth of data that can be evaluated and refined by applications in the field of artificial intelligence. (Martinelli et al., 2021, p. 166) As a result of the Fourth Industrial Revolution, the value of highly skilled human labor will increase, as these machine processes require the supervision of competent personnel who constantly understand how complex and interdependent machine processes work. (Preuveneers and Ilie-Zudor, 2017, p.7) These key technologies and their interdependent modes of operation must, therefore, be promoted across all sectors to equip the economy for the structural competitive conditions of the 21st-century global economy and avoid suffering competitive disadvantages that will be almost impossible to remedy in the long term. The European Gaia-X initiative is the European response to the challenges of the Fourth Industrial Revolution by increasing the competitiveness of the European economy and enabling it to compete with American and Chinese hyperscalers through the promotion of cross-industry data exchange to generate and consolidate the performance of key technologies. This is because the intensification of the Fourth Industrial Revolution is not only the result of technological catalysts but is also driven by a change in the primacy of the organizational form of economic activity from markets to ecosystems, especially since Gaia-X itself is a data infrastructure ecosystem.

4. The characteristics of ecosystems

The Fourth Industrial Revolution is primarily initiated and driven by the expansion and evolution of digital ecosystems such as Gaia-X. In general, an ecosystem can be defined as a set of actors that cooperate to varying degrees to produce multilaterally coordinated and non-generic complementarities, all of which are not subject to a fully hierarchical form of control at any time. (Jacobides et al., 2018, p.2264) The main focus of the definition of a digital ecosystem is that there is an existing and functional coordination structure of value creation so that collectively generated added value can be realized. (Adner, 2017, p.40) Accordingly, its distinguishing characteristics are hierarchy-independent and heterarchy-interdependent structures combined with an existing and functional alignment structure of value creation of the ecosystem. (Autio, 2022, p.99) The long-term success of an ecosystem depends on the recurring and goal-oriented design of three interrelated success conditions: modularity, generativity, and orchestration. The success condition of modularity refers to the decomposition of the ecosystem into modular subsystems, which increases intramodular interdependence and extramodular independence by enabling the individual subsystems to work on idiosyncratic innovations on the one hand and increases the resilience of the entire ecosystem by allowing the individual subsystems to react differently to shocks due to their relative independence on the other hand. Modularity is, therefore, an essential aspect of ecosystem success. (Gawer, 2014, p.1240; Kretschmer et al., 2022, p.406) The success aspect of orchestration refers to the ecosystemic necessity of continuously determining and shaping the relationship between granting and restricting freedoms by an orchestrator to ensure the functioning and success of the ecosystem. Consequently, the challenge of orchestrating the ecosystem is that it is necessary to continuously sound out the appropriate degree of granting and restricting freedoms because too many control measures impair the idiosyncratic, innovative power of the modularly configured subsystems on the one hand. On the other hand, too few control measures jeopardize the cohesion and, thus, the innovative power of the entire ecosystem. (Addo, 2022, p.362; Isckia et al., 2020, p.198) The third aspect of success is generativity. The concept of generativity refers to the ability of ecosystems to produce unanticipated innovations. (Llewellyn Thomas & Tee, 2022, p.270; van der Geest & van Angeren, 2023, p.89) The modular architecture of the ecosystem, in combination with successful orchestration, helps to increase the generativity of ecosystems because the ecosystem participants can organize themselves in the respective ecosystem subsystems and produce innovations that cannot be centrally predetermined and, at the same time, ensure that the ecosystem does not disintegrate over time into numerous increasingly isolated subsystems and that its structure is not entirely and permanently destabilized. In ecosystem research, the aspects of modularity, orchestration, and generativity are often analyzed independently, which is why there is still a great deal of uncertainty about the interrelationships between the concepts. The Gaia-X data infrastructure ecosystem is divided into numerous domains, which can be understood as modular subsystems, and the orchestrator is Gaia-X due to its role in infrastructure provision and technical specifications in continuous operation. Therefore, this paper also undertakes to contribute to a closer integration of the three concepts of modularity, orchestration, and generativity about general ecosystem research by plausibilizing the extension and deepening of the concept of self-descriptions (SD) to self-recommendations, as this is also an essential and Gaia-X overarching issue.

4.1. Strategic information management

The elements of an ecosystem theory just outlined benefit from further development of the theory of strategic information management because the functioning of the diverse intra-organizational and increasingly inter-organizational information flows, which strategic information management examines, also represents an indispensable prerequisite for establishing and operating an ecosystem. This is because inter-organizational collaboration is defined as an intermediate form of market and hierarchical control mechanisms for the design of an ongoing communicative negotiation process in the form of joint cooperation. (Majchrzak et al., 2015 p.1339) The ecosystems are composed of neither market-shaped nor hierarchical ongoing negotiation processes as elementary operational units, which is why both theories are closely related to each other. Strategic information management is an indispensable success factor for modern organizations because it generates competitive advantages through the well-thought-out generation, distribution, and evaluation of information, increasing the organization's competitiveness. (Kannisto et al., 2020, p.655; Holste & Fields, 2010, p.128; Zaheer & Trkman, 2017, p.2) The competitive advantages result from efficiency gains due to the adequate acquisition, evaluation, and optimization of information, which generate insights that make the respective organization more competitive. This is why, given the relevance of information management, it is such an important key competence for organizations in the competitive economic environment of the 21st century. However, strategic information management is currently undergoing major changes across all industries, as the focus on shaping intra-organizational information flows in the context of the emerging ecosystems of the Fourth Industrial Revolution is losing ground to the need to shape inter-organizational information flows. This change of perspective in strategic information management is therefore based on the insight that the intra-organizational generation, distribution, and evaluation of information is increasingly only half the story in the competitive environment of the 21st century because increasing the capacity to absorb new types of inter-organizational information in order to maintain and expand competitiveness is becoming an indispensable prerequisite. (Nooshinfard & Nemati-Anaraki, 2014, p.243) However, this indispensable prerequisite for the inter-organizational exchange of information is problematic because this transformation of strategic information management interlocks organizations more closely with each other, which are thus increasingly in a relationship of both competition and cooperation. The core problem of this intensification of multiple organizational co-opetitive relationships is that, on the one hand, they receive valuable information through the exchange of information that helps them further and, on the other hand, they give up a unique information advantage over the competition through the dissemination of their valuable information. This leads to the main challenge of strategic information management, which is how to implement inter-organizational coordination and control mechanisms at a higher level that address the recurring and inescapable ambivalence between the willingness to share information and the protection of information. (Loebbecke et al., 2016, p.2) Accordingly, research into strategic information management is faced with the challenge of identifying and analyzing the potential for conflict arising from the multiple and diverse co-opetitive relationships and proposing solutions. These co-opetition relationships are inherently conflict-prone as long as the exact causes and effects of conflict potentials are not understood and taken into account. This is particularly annoying because conflict potentials not only occur with regard to co-opetition relationships, but also temporal contradictions of the short-term and long-term nature of cooperation and structural contradictions with regard to the degree of rigidity and flexibility of the form of cooperation are characteristic of the successful design of inter-organizational information management (Castañer and Oliveira, 2020). To make matters worse, little research in the field of strategic information management is currently devoted to inter-organizational conflict potentials, which is out of proportion to the relevance of the topic. (Lumineau et al., 2015 p.43) The increase in the susceptibility of inter-organizational information flows to conflict is also closely related to the increase in the lack of mutual trust due to the added complexity and lack of transparency of the mutual forms of interdependent interdependence. It is, therefore, not surprising that an evaluation of research on the design of intra-organizational and inter-organizational information flows has found that the aspect of trust

receives the most attention in research. (Asrar-ul-Haq & Sadia, 2016, p.7) It can, therefore, be concluded that there is a theoretical gap in the context of information management research in that there are no technically feasible conceptual proposals on how to mitigate the susceptibility to conflict of ubiquitous co-opetition relationships in inter-organizational information exchange processes and how to increase trust among each other. With the concept of self-recommendations (SR), this article attempts to find a solution to contribute to closing the theory gap. With regard to inter-organizational information management, this paper proposes the addition of self-recommendations (SR) to self-descriptions (SD) based on the Gaia-X case study in order to provide ecosystem participants with a differentiated density of information and thus at least partially eliminate the susceptibility to conflict in information exchange processes due to the previous lack of transparency, thereby initializing more efficient information exchange processes. The susceptibility to conflict in co-opetition relationships is mitigated by the fact that the ubiquitously available, more differentiated information situation enables more well-founded and evidence-based decisions to be made about the willingness to exchange information or withhold sensitive information and, as a result, the entire ecosystem and every ecosystem participant can orient themselves and act more successfully. It is, therefore, a general approach that goes beyond the specific requirements and characteristics of the data infrastructure ecosystem, which will be discussed in the next chapter.

4.2. The characteristics of Gaia-X

The Gaia-X project was initialized in 2019 by the German and French Ministers of Economy and pursues the objective of establishing a functional data infrastructure-ecosystem based on compliance with European values, which, as an association of numerous and expandable actors from all industrial sectors, enables and advances data transfer under uniform sovereignty standards and security standards and already represents the most significant European association of companies for the purpose of data exchange. (Autolitano & Pawlowska, 2021, p.13; Bernhardt & Steininger, 2021, p.66) The Gaia-X data infrastructure ecosystem is founded on four fundamental principles. The principle of interoperability ensures that heterogeneous information technology infrastructures of actors are enabled to communicate with each other. In contrast, the principle of portability states that, while adhering to Gaia-X standards, data transfer between participating actors is regulated. The principle of data sovereignty, in turn, ensures that participants are assured of transparency and control over how their data is used in the context of Gaia-X, and the final fourth principle of security and trust clarifies that Gaia-X is entirely secure and trustworthy, both internally and externally. (Rusche, 2022, p.9) Consequently, ensuring the scalability of the principle-driven and principle-regulated transfer of data within data spaces in the context of Gaia-X is central to ensuring Gaia-X's success. Accordingly, it is also central to ensuring the success of Gaia-X that numerous stakeholders recognize added value in participating in the data transfer and are willing to help shape it. (Tardieu, 2022, p.48) Once the Gaia-X data infrastructure-ecosystem is successively perceived as an enabler- through the respective specific matching of providers and consumers of certain data of a variety of industry-specific innovative solutions, then the chances are good that Gaia-X will be a success overall, as in this way existing industry-specific data spaces will flourish. New data spaces will emerge as a result of the constant regenerated interest. (Rusche, 2022, p.19) That said, Gaia-X is not only beneficial for data exchange-based enhancement of corporate competitiveness but also for decentralized advancement of key 21st-century technologies such as Big Data and Artificial Intelligence, as it can be used to aggregate and analyze data volumes that are otherwise only available to the few American and Chinese hyperscalers. (Bernhardt & Steininger, 2021, p.67) The question arises as to how Gaia-X as a decentralized data-infrastructure ecosystem can competitively stand its ground against American and Chinese competition from centralized platform companies, which at the same time raises the question of to what extent and by what means the four Gaia-X principles of interoperability, portability, sovereignty, and security can be given an efficacy that is not only theoretically thought-out and well-sounding but rather empirically manifest. To this end, this paper adopts a data-theoretic starting point by analyzing how data characteristics have fundamentally changed with the transition to the Big Data era and what implications this has for centralized and decentralized-organized platforms. The data-theoretical analysis then makes it possible to highlight what kind of advantages decentralized-organized platforms have over centralized platforms in the treatment and scalability of Big Data and, derived from the entire course of analysis, proposes to extend the self-descriptions of the participants in Gaia-X to include self-recommendations to give the principles of interoperability, portability, sovereignty, and security in Gaia-X the empirically manifesting efficacy they are in need for.

5. The big data era

In general, the Big Data era refers to the transition from a social and economic environment of data scarcity to a social and economic environment of data abundance. (Kitchin, 2013, p.263) The first major catalyst for the emergence and expansion of the Big Data era is the continuous increase in the relevance of data volumes to the economy in that it has successively transformed it to become a data technology-driven and data product-based data economy, as more and more companies have come to view data as an essential strategic resource of the 21st century and are making investments to drive innovation and marketing in a targeted manner, transforming all industries and increasing their productivity to gain a competitive edge (Bany Mohammed et al., 2024; Sakas et al., 2023; Yoshikuni et al., 2023). A second catalyst is the acquisition of consulting expertise in Big Data and related technologies by major consulting firms such as Gartner, McKinsey, Deloitte, PricewaterhouseCoopers, KPMG, and Bloomberg, consulting on industry data issues and the investment of organizations in Big Data Analytics (Kushwaha et al., 2021). Finally, as a third catalyst, states are also competing to invent and apply promising data technologies in their domestic industries to position and compete in a forward-looking manner in the geopolitical and geoeconomic competition between states. (Cao, 2017, p.14f) For this reason, one does not do justice to the complexity of the Big Data era if one reduces it only to the massive increase in data volumes in the various types of industries, as it is equally characterized by an increase in data variety, in that in the form of a triadic sequence of steps, a variety of heterogeneous data sets are generated from a variety of heterogeneous data sources in a variety of heterogeneous economic sectors, which is why the fusion of formerly disparate data sets into a coherent data set represents one of the greatest challenges in the context of the Big Data era. (Lv et al., 2017, p.1891) The societal diffusion of novel technologies, such as the Internet of Things (IoT) or cloud computing in particular, are favoring the emergence of a variety of increasingly heterogeneous data sources from which heterogeneous datasets emerge, the extracted economic benefits of which, in terms of designing new products and services, act as an indispensable catalyst for increasing the competitiveness of companies across all industries. (Jin et al., 2015, p.60) The complication of extracting an economic benefit from linking the disparate datasets into a coherent dataset is that the datasets are generated and available for further processing in varying degrees of structuredness, semi-structuredness, and unstructuredness, significantly complicating their successful and insightful combination. (Yin, 2015, p.144f) Currently, a new phase of the Big Data era is taking place, as smart devices increasingly exchange data with each other in the Internet of Things (IoT), even without human intervention, and consequently significantly shape and regulate this data exchange themselves. (Lee, 2017, p.297f) The Big Data era is commonly perceived as an era of availability and an increase in the amount of data. Still, this characterization falls short, as

the term refers not only to the change in quantity but also to the change in quality. This is because the increase in the approximate completeness of a dataset is synonymous with an increase in the potential correlational richness of the data within the dataset. After all, as the quantity of data increases, the likelihood of finding correlations can only increase. (Chandler, 2015, p.836) Certainly, increasing the amount of data does not necessarily result in increased correlative richness. However, the increase in the amount of data can only have a positive effect on the discoverability of increased correlative richness, as the opposite would mean the absence of the analyzability of vast amounts of data in the form of a small data set, which is always part of the large data set in the age of Big Data anyway. The added correlative richness enables the discovered correlations that were not previously expected but emerged from the data analysis. Although correlations do not shed light on causality, it is increasingly sufficient in a business context to work with robust correlative predictions without paying particular attention to fathoming causality. (Mazzocchi, 2015, p.1251) The Big Data era owes its creative power not only to the dimension of enhancing the self-expressivity of data in the form of massively correlative quantities of data but also to the processual dimension of enabling the development and exploitation of new types of quantification-potentials that could not previously be captured in data form. This processual dimension of the exploitation of novel quantification potentials is called the process of datafication. (Holtzhausen, 2016, p.22) The stabilization and expansion of the increasingly ubiquitously available and analyzable massively correlative quantities of data of the most diverse content stands in a complementary enabling relationship to the development and exploitation of novel data-shaped quantification potentials of the most diverse content. The complementarity of the enabling relationship also applies in the opposite direction because the perpetuation and expansion of ever more heterogeneous quantification-potentials increases the number of available and analyzable massively correlative data of ever more diverse content. (Cao, 2017, p.9) The process of datafication in the sense of advancing quantification of all economic sectors and spheres of life has already had such a large impact that it is considered a natural progression and is thus generalized and normalized. (Dijck, 2014, p.198) However, the characteristic of datafication is not merely limited to the prospective aspect of unlocking and exploiting novel quantification potential. Still, it always includes, as Kenneth Cukier and Viktor-Meyer Schönberger argue, a retractive aspect of shaping quantification activity toward the exploration of specific quantification-potentials based on specific interests. They refer to the mere data transformation of all economic sectors and spheres of life as digitization, whereas the simultaneous transformation of specific quantification-interest on the one hand and the successive exploitation of the quantification-potential of the specific subject area of reality, on the other hand, as – what can be termed- the combination of a prospective aspect and a retractive aspect which mutually characterize the process of datafication in contrast to mere digitization which is based only upon the utilization of the prospective aspect without incorporating the retractive aspect as well. They illustrate their abstract argument with examples. For example, Google's augmented reality glasses quantify people's ways of looking; the social media platform Twitter quantifies users' aberrant thoughts; and the social media platform Linkedin quantifies professional networks. (Cukier & Mayer-Schoenberger, 2013, p.35) For example, the online goods retailer Amazon quantifies users' consumption patterns, and the search engine operator Alphabet quantifies users' search queries. The music streaming service Spotify quantifies the music habits of users, the streaming platform Netflix quantifies the movie habits of users, and finally, the mobility company Tesla quantifies the driving behavior of users. These examples could go on and on. Consequently, the nature of the quantification interest generates the quantification phenomenon, which feeds back to conditions of the possibility of quantification as well as the development of further quantification potential. Therefore, the processual dimension of datafication can be conceived only in its dual form of incorporation of a prospective aspect of the search for quantification-potentials and a retractive aspect of the search for specific quantification-potentials, which, considered in combination, affect the activity of quantification, since from the specificity of the quantification interest emerges the specificity of the quantification potential. The combination of the prospective search for quantification potentials and the retractive quantification interest-driven evaluation of quantification potentials gives the process of datafication its creative power.

Proposition. *Big data means not only an increase in the quantity of data but also a change in the quality of data*

5.1. Big data and artificial intelligence

Both the increase in the process of datafication and the intensification of the granularity of the self-expressivity of the data sets correlate with the available computing power, which makes the processing, linking, and storage of the increasing data sets possible in the first place. Simultaneously, the available computing power correlates with the price for the execution of predictions because the execution of the double dynamic of a continuously increasing available computing power and an accompanying continuous price level reduction influences the prediction costs in such a way that predictions requiring computing power can be executed at an increasingly cheap price level. The continuing reduction in the cost of performing prediction, combined with the variety of application areas, leads to an increasing demand for performing prediction. (Agrawal et al., 2018, p.14) Predictions are also steadily gaining in importance because they are inputs to decisions, as the recognition of intelligent decision options depends on meaningful predictions that establish the space of decision options as the basis for informed decisions. (Hutter, 2009, p.882) Recent advances in artificial intelligence research can be counted as advances in prediction technology because they can be extended and perfected with increasing power and accuracy in prediction based on the ever-decreasing price level for computing power. (Agrawal et al., 2019, p.151f) This has far-reaching implications for the process of datafication and the process of the increase of the self-expressivity of data because it adds a predictive aspect to the retractive and prospective aspect. This can be illustrated with an example. The company Amazon uses artificial intelligence to provide customers with automated product recommendations by analyzing the purchasing behavior of all customers to see what kind of patterns in purchasing behavior can be identified to make a prediction based on which product is best suited for which type of customer. (Kitchin, 2014, p.4) It could be speculatively thought that in the future, Amazon will perfect its predictions based on Artificial Intelligence to such an extent that it could prove more lucrative to send customers unrequested products where the prediction states that the customer would very much like that product, rather than waiting for the relevant customer to order it themselves. (Agrawal et al., 2018, p.16) The initialization and intensification of the predictive capacity of Artificial Intelligence prefigures the directionality of the retractive and prospective aspect of datafication through the ongoing intervention, acting directionally on the specific orientation of the search for the expansion of the granularity of the self-expressivity of the data, thereby producing successes that increase the longing and promise of still further successes in the same directionality. Consequently, the use of AI endows the analysis of Big Data with a preference orientation in the directionality that AI itself generates and that, as a partial self-generation result, feeds back over time to the generation of novel data, making future data itself successively manifestations of the preference orientation.

Proposition. *The use of artificial intelligence increases the predictive power of big data.*

5.2. Capacity advantage and acceleration advantage

The previous reasoning has far-reaching consequences for the

relevance and handling of data disposal rights, which will now be discussed in detail. The increase in AI-based predictive performance that can be extracted from Big Data significantly enhances the requirements for ensuring data security because the damage potentially caused by data theft is becoming more and more serious, and the interest in theft is simultaneously becoming more and more tempting due to the increase in the value of the data volumes. Accordingly, it is only logical that the issue of expanding and ensuring data security is given high priority by digital companies, which have already been affected many times by large-scale cyberattacks. (Schmidt & Cohen, 2013, p.121f) In addition to guaranteeing data security, it is also necessary to ensure the transferability of data disposal rights because all the previously described characteristics of the increase in the self-expressivity of data as well as the characteristics of the process of datafication, particularly with regard to the use of increasingly self-perfecting predictive processes, enhance the ability to understand and act exclusively on the part of those companies that own the corresponding immense amounts of data and the technologies for evaluating them. Kai-Fu Lee declares the age of novel disruptions in basic research on artificial intelligence to be over due to the still unexploited potentials of the most recent disruptions about the transformation into marketable products and services in combination with the unlikelihood of further disruptions in basic research in the near future and accordingly pleads for the design of a double transition between the era of artificial intelligence basic research to an era of comprehensive artificial intelligence implementations and from the era of artificial intelligence expertise to the era of Big Data self-expression. (Lee, 2018, p.28) In contrast to this, Nick Bostrom argues that fundamental disruptions in basic AI research are increasingly likely due to the ever-increasing private sector as well as government competition conditioned by the continued efficacy of the promise of gaining a decisive strategic advantage. (Bostrom, 2017, p.95f). In fact, it follows from the line of reasoning about the Big Data era that the pursuit of the two opposing objectives is not mutually exclusive but rather presupposes each other. The intensifying unfolding of the properties of self-expressivity of Big Data as well as the likewise intensifying prospective, retractive, and predictive aspects of the process of datafication in conjunction with an exclusive right of ownership and disposal persisting in the future course of time provides a decisive strategic advantage both about the strategy of multiplying functional AI implementations and with regard to fundamental research on artificial intelligence, because only on the basis of Big Data is the possibility of realizing a decisive strategic advantage in both directions even thinkable. Consequently, the argument draws attention to the shift in power relations perpetuating power asymmetries in the digital economy and society.

A large number of the most influential and successful companies in the global economy generate enormous amounts of data through platform ownership and concurrently drive the majority of the world's artificial intelligence research with the help of their ever-increasing exclusive data ownership combined with the corresponding purchased human expertise in basic research and further development of artificial intelligence. (Montes & Goertzel, 2019, p.354) These central platform companies benefit from the ongoing and ubiquitous generation of user data, which they continuously use to perfect the performance of algorithms further and thereby successively expand their market power. (Jacobides et al., 2021, p.430) This data-based market power enables the central platform companies to exploit the market power in their interest and against the interests of other stakeholders. (Chen et al., 2021, p.1306) The enhancement of the self-expressivity of Big Data in combination with the expansion possibilities of the prospective, retractive, and predictive aspects of the process of datafication certainly have a favorable effect on the advancing manifestation of a data-driven and artificial-intelligence-forced power asymmetry in favor of the said central platform companies, because the availability of enormous amounts of data represents the initial condition of the conditions of increase in terms of new business fields and profits par excellence. Decentralized platforms currently have the competitive disadvantage of the unavailability of comparable enormous amounts of data to realize the initialization of the initial condition for the realization of the increase of promising new business models. Still, in contrast, they have the advantage of the acceleration of the increase conditions under the provision of the given initial condition, which is why a capacity advantage on the part of the centralized-organized platform companies is diametrically opposed to an acceleration advantage on the part of the decentralized-organized platform companies. This is because, in decentralized-organized platforms, the multiple characteristics of the numerous platform participants can manifest themselves more strongly than is the case with respect to centralized platforms. After all, they act as equal-ranking contractual partners and not as unilateral service recipients. (Goertzel et al., 2017, p.67f) Due to the advantage of the multiple specificities of the platform participants in decentralized platforms, they can make use of the mechanism of challenge propagation by finding the appropriate specific partners for specific problems in a platform-mediated way who are capable of solving the problem. (Heylighen, 2017, p.77) Thus, if the dual mechanism of unfolding the multiple specificities of decentralized-organized platform companies in combination with a multiplication of communicated challenges and challenge solutions can manifest on a decentralized-organized platform, then this results in the emergence of the givenness initial condition of the availability of Big Data in combination with the acceleration advantage of the expansion of Big Data over centralized-organized platforms. The realization of this dual mechanism, in turn, fundamentally requires, on the one hand, a solution as to how comprehensive and willing cooperation among the organizations can be initially established and, on the other hand, should this be successful, the related solution as to how comprehensive, willing and successful cooperation among the organizations can be maintained over time. (Last, 2017, p.51)

Proposition. *Decentralized platforms have a capacity advantage, while centralized platforms have a speed advantage.*

6. Self-descriptions (SD) and self-recommendations (SR)

Consequently, this problem also arises for the Gaia-X data infrastructure ecosystem. The findings on the acceleration advantage of decentralized platforms and the capacity advantage of centralized platforms are related to Gaia-X in the sense of the question posed in this paper. For this purpose, however, it is first necessary to address the elementary concept of self-descriptions (SD) in the Gaia-X context. Gaia-X self-descriptions (SD) describe entities of the Gaia-X Conceptual Model in a machine-interpretable format. The entities described in the form of Self-Descriptions are Gaia-X Participants as well as the resources and service offerings of providers. When creating and implementing self-descriptions (SD), the most important thing is that they follow a well-defined Self-Description schema, guaranteeing a unified representation of the Self-Description so that they successfully fulfill their function of establishing and ensuring the discovery and comparison of entities within Gaia-X. Combined with trusted verification mechanisms, the well-defined self-descriptions (SD) empower Gaia-X Participants to initiate well-structured decision-making processes, as they are provided with the appropriate information that enables straightforward discoverability and structured comparability of offerings for specific interest situations. The Gaia-X participants, and especially the providers, are responsible for creating self-declared information about themselves as well as about their offers. In addition, self-descriptions (SD) consist of a set of Verifiable Credentials, which in turn consist of a set of Self-Claims. Different trusted parties within Gaia-X sign the Verifiable Credentials. The set of all Self-Descriptions, in conjunction with all relations them to each other, forms the active lifecycle state and is referred to as the Self-Description Graph, where the Self-Descriptions form the nodes and the relations among the Self-Descriptions form the edges. (Gaia-X, 2023, p.20f) At this point, it is worth transferring the data-theoretical findings

on the characteristics of Big Data and the characteristics of the process of datafication in relation to centralized-organized platforms and decentralized-organized platforms to Gaia-X in the form of a capacity advantage for the former over an acceleration advantage for the latter. Suppose one transfers the collected findings of the entire line of reasoning to the question of the conditions for the success of Gaia-X. In that case, it becomes evident that it is promising if the Self-Descriptions (SD) already firmly anchored in Gaia-X are extended by Self-Recommendations (SR), which, in the sense of their functionality of creating well-structured decision-making processes, extend the self-declared information about themselves and about their offerings to include self-declared insights into the results of their offerings as well as the resulting options for other Gaia-X participants in the form of self-recommendations (SR). In the form of self-recommendations (SR), dataset providers would consequently not only write descriptions of the dataset's characteristics but also include insights on the dataset's actual performance in the form of, for example, statistical analyses so that the interested party is provided with more profound information that substantiates and expands the well-structured decision-making process beyond the performance of mere self-descriptions (SD). The more in-depth information anchored in the Self-Recommendations about the provider datasets helps the interested Gaia-X Participants to make better decisions due to the givenness of better information. The integration of Self-Recommendations into the Gaia-X framework would thus increase the performance of decision-making processes compared to the variant of a one-sided use of Self-Descriptions (SD), whereby Gaia-X as a decentralized-organized data infrastructure-ecosystem can, at the same time express the acceleration advantage derived above more and more and increasingly better over time.

Proposition. *Adding self-recommendations to self-descriptions is advantageous due to the increased amount of information for decision-making.*

As Kebede (2010) describes, increasing knowledge is the ultimate goal of information science. This is where the idea of self-recommendations comes in: knowledge about data and services is expanded to include possible applications, uses, and processing options. In this way, the additional information provided by SR, following the doctrine of information management, contributes to a positive change in how people and organizations deal with data (International Journal of Information, Management). For example, SR can lead to more efficient data analysis because it provides recommendations on handling the data and services in question rather than simply describing them. As Edwards (2022) explains, information and knowledge about handling it are inextricably linked, underscoring the importance of the conveyed knowledge. In addition, this knowledge provides the basis for the discourse between the exchanging parties, which is necessary for the further use of the data. According to Galliers et al. (2020), this discourse is essential to ensure creativity and innovation in the context of exchange. Existing information and communication technologies can support some of this dialogue but not the transfer of experience and knowledge. SR, therefore, extends the capabilities of existing ICT systems to include interaction and exchange of understanding between two parties who are not familiar with each other.

In addition to increasing the amount of knowledge transferred and thus improving the exchange, SRs can positively impact information management strategy formation (See Fig. 1, Own Illustration, based on Afifi and Weiner (2004)). As Afifi and Weiner (2004) describe in their Theory of Motivated Information Management (TMIM), uncertainty and discrepancies in the interpretation phase can lead to anxiety for the information procurer and, thus, poorer results and reduced effectiveness in the evaluation phase. This has a negative impact on the chosen information management strategy in the decision phase. This is exacerbated by the iterative approach within TMIM, where decisions made in the decision phase influence future assessments in the evaluation phase. This is where SR comes in with its additional information. The extra knowledge reduces uncertainty and discrepancies in the interpretation

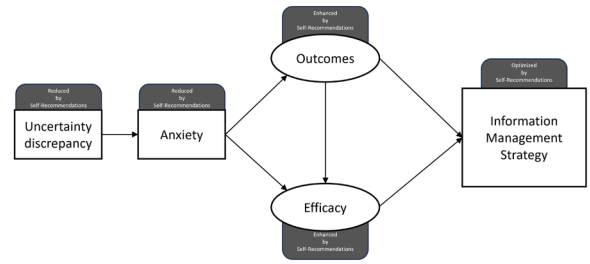

Fig. 1. Theory of Motivated Information Management improved by SR (Own illustration, simplified, based on Afifi & Weiner, 2004).

phase and can, therefore, reduce or eliminate potential anxiety. According to TMIM, this positively impacts the results and effectiveness achieved in the evaluation phase and, thus, the information management strategy developed.

The importance of SR for developing a successful information strategy is further emphasized by Reponen (1993). He shows that information management decisions are becoming more economic, business-oriented, and less technical. Therefore, managers need to understand the role of data and information technology and be actively involved in the decision-making process to develop a successful information management strategy. According to Reponen, this requires a clear understanding of data and technology. Self-recommendations positively contribute to this, as they contain technical, business potential, and explanations, which can help decision-makers understand their usability. In this way, managers can understand the added value of a data set without in-depth technical expertise, thus reducing their dependence on IT experts in their decision-making. As a result, all business and strategic decisions can be made more efficiently and informed, positively impacting business results. Thus, SR provides data analytics benefits to technicians through increased transparency and a business decision advantage by revealing the business value of data and services.

7. Discussion

This discussion chapter outlines this paper's contribution to the literature and the practical implications that SR can have. The research literature contributes to introducing an innovative concept for promoting data exchange processes in decentralized ecosystems, which can be investigated even more comprehensively and empirically in further research. The practical implications, in turn, range from the influences on data handling and the associated improvements in efficiency to reductions in transaction costs and increases in innovation.

7.1. Contribution to literature

With self-recommendations (SR), future research has a new concept at its disposal and, building on this research, can empirically investigate whether the introduction of self-recommendations (SR) as an extension of self-descriptions (SD) contributes to the optimisation of incentives for participation in the data exchange processes and how strong the effect is. In addition, research can be conducted into exactly what type of key figures the self-recommendations (SR) should be composed of so that the data exchange processes can be optimally supported on the part of the data providers and the data consumers. In addition, future research can also build on these research results to investigate the influence of the introduction of self-recommendations (SR) on pricing in decentralised ecosystems by ensuring that more sound information on the quality of the data is available. Ultimately, this research represents a significant further step towards a comprehensive theory of decentralised ecosystems that has yet to be developed, because such a theory does not yet exist, and a series of further fundamental research will be necessary in

order to successfully conceptualise such a theory. In this respect, this article also makes a scientific contribution to understanding the functioning and maximising the prospects of success of decentralized organised ecosystems in general.

7.2. Practical implications

The practical implications of the proposed Self-Recommendations are manifold. Firstly, there is a positive impact on the cost efficiency achieved by improving data usability. Data usability is improved to the extent that relevant data and services can be identified more quickly and accurately by users, thereby increasing selection performance and making data management more efficient. This reduces transaction costs and bad investments in unsuitable data sets and services. The SR's specific application recommendations can also drive the development of best practices and cross-industry standards, which would be particularly beneficial in sectors where data integration is a major challenge. Companies can also more easily identify potential synergies and drive innovation and collaboration based on the recommended data usage guidelines. Finally, the optimized use of resources represents a practical implication. Through precise recommendations, resources such as storage space and computing capacity can be used more efficiently as users can better identify which data and services are most valuable for their needs. With regard to the real-time practicability of SR, it should be mentioned that a technical implementation should take place so that SR can be seamlessly integrated into existing systems (e.g., by means of an API). Using automated systems and machine learning, SR could be continuously improved based on usage data and feedback. Such systems could recognize patterns and make recommendations more effective. This effectiveness also depends heavily on how intuitive and understandable the SRs are for the user. Real-time dashboards or user interfaces could be developed here to present the SRs understandably and support users in making real-time decisions.

8. Conclusion

The success of the European decentralized Gaia-X data infrastructure-ecosystem requires ensuring and intensifying a successful data exchange between data providers and data consumers so that numerous data-based innovative products and services can be created in the multiple and between the multiple Gaia-X Domains and the competitiveness of European companies can be increased across industries. Should this goal not be achieved, European companies are threatened with an expanding dependence on American and Chinese centralized-organized platform companies. Therefore, it is essential for the future competitiveness of the European economy that the decentralized-organized data infrastructure-ecosystem Gaia-X becomes successful. For this reason, the paper has investigated from a data-theoretic perspective how the Big Data era affects the design of centralized-organized and decentralized-organized platforms and what implications can be derived for the Gaia-X decentralized-organized data infrastructure-ecosystem.

It could be derived data-theoretically that the decentralized-organized data infrastructure ecosystem Gaia-X can express its acceleration advantage over the capacity advantage of the centralized-organized platforms if the self-descriptions (SD) anchored in Gaia-X are extended by so-called self-recommendations (SR) because this is a much more effective way of ensuring that data exchange between data providers and data consumers takes place on Gaia-X in general and in the Gaia-X Domains in particular and that it intensifies over time. This is because self-recommendations (SR) enhance the power of Self-Descriptions by enabling providers to not only provide self-declared information about their data but also to provide additional self-declared insights into the results of their data offerings, enabling data consumers to make more informed decisions as they are much better able to assess the value of the data.

Therefore, based on the previous argumentation, the first recommendation is to complement the existing self-descriptions (SD) in the Gaia-X ecosystem with self-recommendations (SR) to increase the willingness of ecosystem participants to participate in data exchange processes. The second recommendation for action is to design the self-recommendations (SR) so that the added value of the data performance is recognizable beyond the mere description of the type of data set. Of course, machine readability and the requirements for self-descriptions must be considered. The third recommendation is for scientific research. This paper's results show a great need for complementary economic research on the Gaia-X ecosystem, as it has been considered almost exclusively from a technical perspective. Filling this research gap can help position the decentralized European approach competitively against competing models from the US and China by building on the idiosyncratic strengths of the decentralized European approach.

CRediT authorship contribution statement

Vincent Geilenberg: Writing – review & editing, Writing – original draft. **Wolfgang H. Schulz:** Writing – review & editing, Writing – original draft. **Jonathan Mize:** Writing – review & editing. **Henrik Kleis:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors report that there are no competing interests to declare.

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